

JEE MAIN REVISION MASTER: HYDROGEN OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 86 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Hydrogen - Core Application	86 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrogen - Core Application (Very High Yield — 86 PYQs)

Core Principle: Key problem pattern in Hydrogen.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Isotopes of hydrogen hav..." [Ans: B]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: Statement-I : Methane and steam passed over a heated Ni catalyst produces hydrogen gas. Statement-II : Sod..." [Ans: C]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Loss of electron from hy..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 13th April Evening Shift (Q#1)

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Isotopes of hydrogen have almost same chemical properties, but difference in their rates of reaction. Reason R : Isotopes of hydrogen have different enthalpy of bond dissociation. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) is not correct but R is correct
- (B) Both A and R are correct but R is NOT the correct explanation of A [**Correct**]
- (C) is correct but R is not correct
- (D) Both A and R are correct and R is the correct explanation of A

Correct Answer: B

Solution Analysis: Assertion A : Isotopes of hydrogen (protium, deuterium, and tritium) have almost the same chemical properties because they have the same number of protons and electrons, which dictate chemical behavior. However, they differ in their rates of reaction due to differences in their atomic masses, which affect reaction kinetics. Reason R : The isotopes of hydrogen have different enthalpy of bond dissociation due to their differences in atomic mass. The stronger bond in deuterium compared to protium, for example, leads to higher bond dissociation enthalpy. This, in turn, influences the rate of react